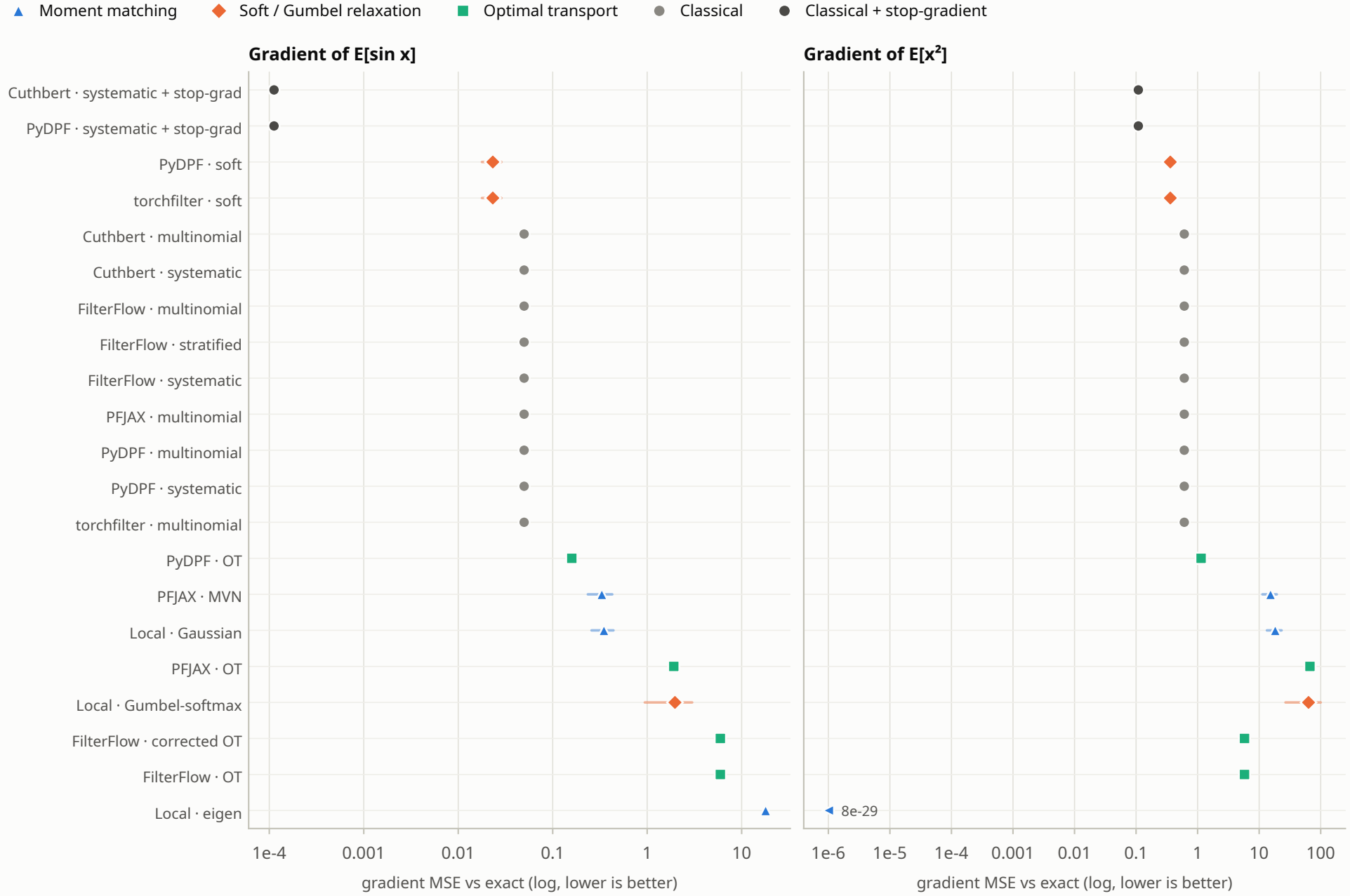
Local expectation gradients through one resampling step

Gaussian-mixture cloud · N = 128 · reference is the exact derivative of the weighted discrete distribution



Source: gradient_summary.csv. Values below 1e-6 are drawn at the left edge with their value; eigen matches x^2 gradients by construction (moment matching).

Classical resamplers without stop-gradient, and OT, return the same gradient for every seed here (zero variance), so their error is pure bias.